

Round-off Error Instability

①

Derivative formulas we have derived in last lecture are affected by the round off error due to h being in the denominator.

Consider three point midpoint formula

$$f'(x_0) = \frac{1}{2h} [f(x_0+h) - f(x_0-h)] - \frac{h^2}{6} f'''(\xi_0) \quad \text{--- ①}$$

Small value
of h will increase
round off error

Small value of
 h will decrease
truncation error

We are interested in the optimal value of h in a given interval that ~~reduces~~ reduces the total error.

Suppose $f(x_0+h) = \tilde{f}(x_0+h) + e(x_0+h)$

Exact
value

approximate
value

round-off error

and

$$f(x_0-h) = \tilde{f}(x_0-h) + e(x_0-h)$$

Substitute these values in ①

$$f'(x_0) = \frac{1}{2h} \left[\tilde{f}(x_0+h) + e(x_0+h) - \tilde{f}(x_0-h) - e(x_0-h) \right] - \frac{h^2}{6} f'''(\xi_0)$$

$$\underbrace{f'(x_0)}_{\text{Exact}} - \underbrace{\frac{1}{2h} [\tilde{f}(x_0+h) - \tilde{f}(x_0-h)]}_{\text{approximate}} = \underbrace{\frac{1}{2h} [e(x_0+h) - e(x_0-h)]}_{\text{round off Error}} - \underbrace{\frac{h^2}{6} f'''(\xi_0)}_{\text{truncation error}}$$

Error

$$\text{Total error } E = \frac{1}{2h} [e(x_0+h) - e(x_0-h)] - \frac{h^2}{6} f^{(3)}(\xi_0) \quad (2)$$

We assume that $|e(x_0 \pm h)| < \varepsilon$ i.e. maximum round off error is bounded by ε and $|f^{(3)}(\xi_0)| < M$ for $x_0 - h < \xi_0 < x_0 + h$,

Then

$$\begin{aligned} |E| &= \frac{1}{2h} |e(x_0+h)| + \frac{1}{2h} |e(x_0-h)| + \frac{h^2}{6} |f^{(3)}(\xi_0)| \\ &\leq \frac{1}{2h} \varepsilon + \frac{1}{2h} \varepsilon + \frac{h^2}{6} M \end{aligned}$$

$$|E| \leq \frac{\varepsilon}{2h} + \frac{h^2}{6} M$$

let us assume, $g(h) = \frac{\varepsilon}{h} + \frac{h^2}{6} M$ and find h

that minimizes g .

$$g'(h) = -\frac{\varepsilon}{h^2} + \frac{h}{3} M$$

$$g'(h) = 0 \Rightarrow \frac{\varepsilon}{h^2} = \frac{h}{3} M, \quad h = \sqrt[3]{\frac{3\varepsilon}{M}}$$

$$h = \left(\frac{3\varepsilon}{M} \right)^{1/3}$$

$$g''(h) = \frac{2\varepsilon}{h^3} + \frac{M}{3}$$

$g''\left(\left(\frac{3\varepsilon}{M}\right)^{1/3}\right) > 0$ Therefore $h = \left(\frac{3\varepsilon}{M}\right)^{1/3}$ minimizes $g(h)$.

Q No 9 Consider the following table.

③

x	f(x)
-3.0	9.36
-2.8	8.23
-2.6	7.180
-2.4	6.209
-2.2	5.320
-2.0	4.513

This data is taken from $f(x) = e^{x/3} + x^2$.

① From the data find $f'(-2.6)$ using all possible values of h in ~~the~~ Three-Point Midpoint formula.

~~Ex. $x_0 = -2.6$~~
sol

For $x_0 = -2.6$, there are two possible values of

h ; ~~$h = 0.2$ or $h = 0.4$~~

$$h = 0.2$$

$$f'(-2.6) = \frac{1}{2(0.2)} [f(-2.4) - f(-2.8)] = \frac{1}{2(0.2)} [6.209 + 8.23]$$

$$h = 0.4$$

$$f'(-2.6) = \frac{1}{2(0.4)} [f(-2.2) - f(-3.0)] = \frac{1}{0.8} [5.320 - 9.36]$$

② If $\epsilon = 10^{-4}$, find the best value of h

using $h = \left(\frac{3\epsilon}{M} \right)^{1/3}$.

④

~~$$M = M_K \left| f^3 \left(\frac{p}{K} \right) \right|$$~~

$$f'(x) = \frac{1}{3} e^{x/3} + 2x$$

$$f''(x) = \frac{1}{9} e^{x/3} + 2$$

$$f'''(x) = \frac{1}{9} e^{x/3}, \quad f'''(\varphi(x)) = \frac{1}{9} e^{\varphi(x)/3}$$

~~Q.1~~ The given data is in range $[-3.0, -2.0]$.

hence for ~~the p.d.f~~ $-3.6 < f(x) < 2.6$

$\therefore f'''(g(x)) = \frac{1}{9} e^{g(x)/3}$ is an increasing function.

~~and~~ ~~not x~~ ~~not~~ ~~f~~ ~~(-2.3)~~

$$|f'''(x)| / \leq f'''(-2.0) = \frac{1}{9} e^{-2.0/3} = 0.6570$$

$$M = 0.0570$$

$$h = \left(\frac{3(10^{-4})}{0.0570} \right)^{1/3} = 0.1739$$

$h \approx 0.2$ is the best choice from the table.